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DOCUMENT 3 — FUEL CELL

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Component: Hydrogen Fuel Cell

Purpose:

Convert stored hydrogen into electricity and useful heat.

Inputs:

- Hydrogen mass flow m_{H2_dot} [kg/s]

Outputs:

- Electrical power P_{el_out} [W]
- Heat flow Q_{heat} [W]
- Heat output temperature level T_{heat_K} [K]

Key Variables:

η_{fc} : electrical efficiency [-]

T_{heat_K} : heat output temperature level [K]

Equations:

$P_{el_out} = \eta_{fc} * m_{H2_dot} * LHV_{H2}$

$Q_{heat} = (1 - \eta_{fc}) * m_{H2_dot} * LHV_{H2}$

$\eta_{fc} = \eta_{fc}$

$T_{heat_K} = T_{heat_K}$

Assumptions:

- Constant efficiency assumed
- System-level energy balance applied